

LLM Agent

From simple meaning to architecture and examples

Learning approach

This PDF starts with a very simple explanation first. Then it gradually increases the depth with components, architecture, examples, use cases, and implementation flow.

1. First: very simple explanation

Simple definition

An agent in LLM means an AI system that can think, decide, use tools, and take actions to complete a task.

A normal LLM usually works like this:

Normal LLM flow

User asks -> LLM answers

An LLM agent does more than just answer:

LLM agent flow

User gives goal -> Agent plans -> Uses tools/data -> Takes steps -> Checks result -> Final answer

Simple example

User asks: "Find the latest price of iPhone and compare it with Samsung."

A normal LLM may only give a general answer. An agent can:

1. Search the web
2. Open product pages
3. Extract prices
4. Compare specifications
5. Prepare a summary
6. Give the final recommendation

In simple words

An LLM agent is like a smart assistant that can act on behalf of the user.

Component	Meaning
LLM	Brain of the agent
Tools	Calculator, web search, database, API, files
Memory	Remembers previous information
Planning	Breaks a big task into small steps
Action	Executes tasks using tools
Observation	Checks tool results and decides next step

Example in corporate training

Suppose you ask: “Analyze this PDF and create a quiz.”

An LLM agent can read the PDF, understand topics, generate questions, create answers, and export the quiz to Excel or PDF.

Normal LLM	LLM Agent
Gives text response	Can perform tasks
Mostly one-step answer	Multi-step reasoning
Cannot directly use tools unless connected	Uses tools/APIs/files/search
Good for explanation	Good for automation

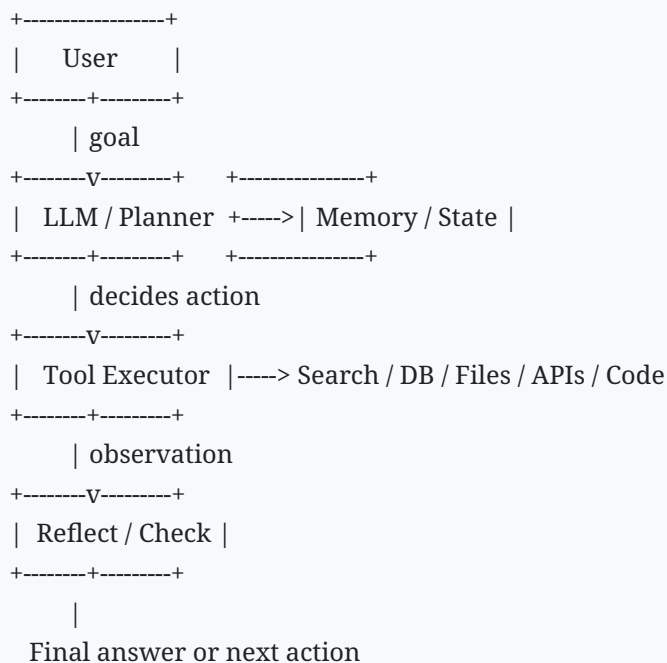
Real-time examples: customer support agent, travel booking assistant, code debugging agent, data analysis agent, resume screening agent, RAG document Q&A agent, and email scheduling assistant.

2. Gradually deeper: what makes an LLM agent powerful?

An LLM agent is not only a model. It is a small system around the model. The model understands the goal, but tools, memory, planning, and feedback loops help it complete real work.

Layer	Purpose	Example
User goal	The task the user wants completed	Create a 20-question quiz from this PDF
Reasoning / planning	Decide the sequence of steps	Read PDF -> identify topics -> create questions -> review
Tool use	Interact with outside systems	PDF reader, calculator, database, search, email
Memory	Keep useful context across steps	Remember the selected document and target audience
Observation	Read tool output and decide next action	Tool found 8 topics; now create questions
Final response	Return the completed result	Quiz, answer key, exported file

Typical LLM Agent Architecture



3. Agent loop: plan, act, observe, improve

7. Understand the user goal and constraints.
8. Break the goal into smaller tasks.
9. Select the right tool for the next step.
10. Execute the tool or action.
11. Observe the result.
12. Check whether the result is enough or whether another step is needed.

13. Return the final output in a useful format.

Important idea

The agent does not need to know everything internally. It can use tools to get information, perform calculations, read files, query databases, or create outputs.

4. Types of LLM agents

Agent type	What it does	Example
Tool-using agent	Uses external tools when needed	Calculator, web search, Python, SQL
RAG agent	Retrieves relevant documents before answering	Company policy Q&A bot
Workflow agent	Follows a predefined process	Invoice processing workflow
Autonomous agent	Plans and executes many steps with less human guidance	Research assistant
Coding agent	Writes, tests, and fixes code	Debug a Python Streamlit app
Data analysis agent	Loads data, analyzes it, and creates insights	Analyze sales CSV and create charts

5. More examples

Use case	How the agent works
Customer support	Reads complaint, checks knowledge base, suggests response, creates ticket summary
Travel planning	Searches flights/hotels, compares options, prepares itinerary
Training content creation	Reads course goal, creates outline, examples, exercises, quizzes
Data analysis	Reads dataset, cleans data, creates charts, explains insights
Code debugging	Reads error, inspects code, identifies cause, suggests corrected code
HR screening	Reads resumes, compares with job description, prepares shortlist

6. Agent in RAG systems

In a normal RAG system, the application retrieves documents and sends them to the LLM. In an agentic RAG system, the agent decides what to retrieve, whether to search again, how to verify the answer, and when to stop.

Agentic RAG Flow

Question -> Agent decides search query -> Retriever -> Documents
 Documents -> Agent checks relevance -> Search again if needed

Relevant evidence -> LLM answer -> Verification -> Final answer

7. Benefits and challenges

Benefits	Challenges
Can automate multi-step work	Can become complex to design
Can use fresh data through tools	Tool errors must be handled
Can work with files, APIs, and databases	Needs guardrails and validation
Can improve productivity	Cost and latency may increase
Can personalize output using memory	Memory must be controlled carefully

8. Simple implementation pattern

14. Define the agent goal clearly.
15. List the tools the agent is allowed to use.
16. Write instructions for when each tool should be used.
17. Store intermediate state or memory if needed.
18. Add validation so the agent checks its answer.
19. Log every step for debugging and audit.
20. Return output in the required format such as text, table, PDF, Excel, or API response.

Final simple definition

An LLM agent is an AI assistant powered by an LLM that can plan, use tools, and take actions to achieve a goal.